

SystemC + MSVC Complete Setup Guide

This document explains: 1. Building and installing SystemC using MSVC 2. Creating a SystemC Visual Studio project 3. Building and running a SystemC application 4. Required MSVC-specific settings (/vmg)

Part 1: Build and Install SystemC using MSVC

Open:

x64 Native Tools Command Prompt for VS 2022

Verify tools:

```
cl
cmake --version
```

You should see MSVC compiler information.

Assume SystemC source is located at:

```
D:\Practive_Learning\SystemC\systemc
```

Create build folder:

```
D:
cd D:\Practive_Learning\SystemC\systemc

mkdir build_msvc
cd build_msvc
```

Configure SystemC:

```
cmake .. ^
-G "Visual Studio 17 2022" ^
-A x64 ^
-DCMAKE_CXX_STANDARD=17 ^
-DCMAKE_INSTALL_PREFIX=../install_msvc
```

Build Release:

```
cmake --build . --config Release
```

Install:

```
cmake --install . --config Release
```

After installation:

```
D:\Practive_Learning\SystemC\systemc\install_msvc
```

Expected folders:

```
include  
lib  
share
```

Part 2: Create Visual Studio SystemC Project

Open Visual Studio.

Select:

File → New → Project → Empty C++ Project

Choose:

- Platform: x64
- Configuration: Release (or Debug)

Add source files:

```
main.cpp  
i2c_controller.cpp  
i2c_controller.h  
i2c_registers.cpp  
i2c_registers.h
```

Part 3: Configure Project Settings

Open:

Project → Properties

Select:

All Configurations x64

Include path

C/C++ → General → Additional Include Directories

Add:

D:\Practive_Learning\SystemC\systemc\install_msvc\include

Library path

Linker → General → Additional Library Directories

Add:

D:\Practive_Learning\SystemC\systemc\install_msvc\lib

Add SystemC library

Linker → Input → Additional Dependencies

Add:

systemc.lib

Set C++ Standard

C/C++ → Language → C++ Language Standard

Select:

ISO C++17 (/std:c++17)

Runtime Library

C/C++ → Code Generation → Runtime Library

Release:

Multi-threaded DLL (/MD)

Debug:

Multi-threaded Debug DLL (/MDd)

Make sure Debug and Release match how SystemC was built.

Critical SystemC Setting

Open:

C/C++ → Command Line → Additional Options

Add:

/VMG

(or lowercase /vmg)

This flag is extremely important.

SystemC internally uses member-function pointers for:

- SC_METHOD
- SC_THREAD
- callbacks
- process registration

Without /vmg a manually created Visual Studio project may:

- compile successfully
- link successfully
- crash during execution

Typical crash:

```
0xC0000005
Access violation
0xFFFFFFFFFFFFFFFF
```

CMake-generated MSVC projects usually add the required behavior automatically, but manual projects may require this setting.

Part 4: Build and Run

Build:

Build → Build Solution

or:

Ctrl + Shift + B

Run:

Debug → Start Without Debugging

or:

Ctrl + F5

Expected:

```
SystemC banner  
Constructor called  
SC_METHOD executed
```

Important Notes

Keep GCC and MSVC installations separate.

Example:

```
build_gcc  
install_gcc  
  
build_msvc  
install_msvc
```

Never mix:

- install_gcc with Visual Studio projects
- libsystemc.a with MSVC
- systemc.lib with GCC

Use:

GCC → libsystemc.a MSVC → systemc.lib